第十讲 (2026.10.8)

Def $(X, \langle \cdot, \cdot \rangle)$

$\{e_\alpha\}_{\alpha \in I}$ —— O.N.S.

如果 $\forall x \in X$

$$x = \sum_{\alpha \in I} \langle x, e_\alpha \rangle e_\alpha$$

则称 $\{e_\alpha\}_{\alpha \in I}$ 为 X 的一个 O.N.B.

Bessel 不等式: $\forall x \in X$

$$\sum_{\alpha \in I} |\langle x, e_\alpha \rangle|^2 \leq \|x\|^2$$

注

$$\tilde{I} \overset{\text{def}}{=} \{\alpha \in I : \langle x, e_\alpha \rangle \neq 0\}$$ 至多可数

$\tilde{I} = \{\alpha_k\}_{k=1}^{\infty}$

$$\Longrightarrow \sum_{\alpha \in I} |\langle x, e_\alpha \rangle|^2 \overset{\text{def}}{=} \sum_{k=1}^{\infty} |\langle x, e_{\alpha_k} \rangle|^2$$

Q: $$\sum_{\alpha \in I} \langle x, e_\alpha \rangle e_\alpha = \sum_{k=1}^{\infty} \langle x, e_{\alpha_k} \rangle e_{\alpha_k} \ ?$$

i.e. 1° RHS 是否收敛?

2° 对 $\tilde{I}$ 的不同排列, RHS 是否相同?

Lem H — Hilbert space.

$\{e_k\}_{k=1}^{\infty}$ — O.N.S.

$M \overset{\text{def}}{=} \overline{\text{Span}\{e_k\}_{k=1}^{\infty}}$

ii) $\forall x \in H$,

$$\sum_{k=1}^{\infty} \langle x, e_k \rangle e_k = P_M x$$

Pf ⇐ Bessel

$$\sum_{k=1}^{\infty} |\langle x, e_k \rangle|^2 \leq \|x\|^2$$

$$\Rightarrow \left\| \sum_{k=n}^{m} \langle x, e_k \rangle e_k \right\|^2 = \sum_{k=n}^{m} |\langle x, e_k \rangle|^2$$

$$\to 0 \quad \text{as} \quad m, n \to \infty$$

$$\Rightarrow \left\{ \sum_{k=1}^{n} \langle x, e_k \rangle e_k \right\}_{n=1}^{\infty}$$ 收斂 (H 中基本列.)

$$\Rightarrow \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k = \lim_{n \to \infty} \sum_{k=1}^{n} \langle x, e_k \rangle e_k \in H$$

(由) $\langle x - \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k , e_m \rangle = \langle x, e_m \rangle - \langle x, e_m \rangle$

$$= 0 \quad . \quad \forall m$$

$$\Rightarrow x - \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k \in M^{\perp}$$

$$\Rightarrow \quad \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k = P_M x$$

$$(\because \quad H = M \oplus M^{\perp})$$

Cor 对 $\mathbb{N}$ 上 ∞ 阶一置换 σ

$$\sum_{k=1}^{\infty} \langle x, e_{\sigma(k)} \rangle e_{\sigma(k)} = \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k$$

Pf: $\Leftarrow$

$$M \overset{\text{def}}{=} \overline{\operatorname{Span}\{e_k\}_{k=1}^{\infty}}$$

$$\tilde{M} \overset{\text{def}}{=} \overline{\operatorname{Span}\{e_{\sigma(k)}\}_{k=1}^{\infty}}$$

$\Rightarrow$)

$$M = \tilde{M}$$

$$\Rightarrow \quad LHS = P_{\tilde{M}} x = P_M x = RHS$$

Thm H —— Hilbert space

$\{e_\alpha\}_{\alpha \in I}$ —— O.N.S.

$\forall\, x \in H$,

(i) $\sum_{\alpha \in I} \langle x, e_\alpha \rangle e_\alpha \in H$.

(ii) $\left\| x - \sum_{\alpha \in I} \langle x, e_\alpha \rangle e_\alpha \right\|^2 = \|x\|^2 - \sum_{\alpha \in I} |\langle x, e_\alpha \rangle|^2$

Pf $i\overset{A}{\Rightarrow}$

- $$\left\{ \alpha \in I : \ \langle x, e_\alpha \rangle \neq 0 \right\} = \left\{ \alpha_k \right\}_{k=1}^{\infty}$$

Lem + Cor

$$\Longrightarrow \quad \sum_{\alpha \in I} \langle x, e_\alpha \rangle e_\alpha = \sum_{k=1}^{\infty} \langle x, e_{\alpha_k} \rangle e_{\alpha_k} \in H$$

$\forall n$

$$\left\| x - \sum_{k=1}^{n} \langle x, e_{\alpha_k} \rangle e_{\alpha_k} \right\|^2 = \|x\|^2 - \sum_{k=1}^{n} \left| \langle x, e_{\alpha_k} \rangle \right.$$

- $n \to \infty$

$$\Longrightarrow \quad \left\| x - \sum_{k=1}^{\infty} \langle x, e_{\alpha_k} \rangle e_{\alpha_k} \right\|^2 = \|x\|^2 - \sum_{k=1}^{\infty} \left| \langle x, e_{\alpha_k} \rangle \right|^2$$

Thm: H —— Hilbert space

$\{e_\alpha\}_{\alpha \in I}$ —— O.N.S.

- 以下三条等价，

(i) $\{e_\alpha\}_{\alpha \in I}$ 是 O.N.B.

(ii) $\forall x \in H$

$$\|x\|^2 = \sum_{\alpha \in I} |\langle x, e_\alpha \rangle|^2 \quad \left(\begin{array}{l} \text{Parseval} \\ \text{Identity} \end{array} \right.$$

- (iii) $\{e_\alpha\}_{\alpha \in I}$ 完备. i.e. $\left(\{e_\alpha\}_{\alpha \in I} \right)^{\perp} = \{0\}$.

Pf 1° (i) $\Longrightarrow$ (ii)

由 ii - Thm

2° (ii) $\Longrightarrow$ (iii)

假设 $(\{e_\alpha\}_{\alpha\in I})^\perp \neq \{0\}$

$\Longrightarrow$ $\exists\ 0 \neq x_0 \in H$ s.t.

$$\langle x_0, e_\alpha \rangle = 0, \quad \forall \alpha \in I$$

$$\Longrightarrow \sum_{\alpha\in I} |\langle x_0, e_\alpha\rangle|^2 = 0$$

P.I. $\Longrightarrow$ $\|x_0\|^2 = 0$, 矛盾

3° (iii) $\Longrightarrow$ (i)

假设 $\{e_\alpha\}_{\alpha\in I}$ 不是 O.N.B.

$\Longrightarrow$ $\exists\ x_0 \in H$ s.t.

$$x_0 - \sum_{\alpha\in I} \langle x_0, e_\alpha\rangle e_\alpha \neq 0$$

$\forall \beta \in I$,

$$\left\langle x_0 - \sum_{\alpha\in I} \langle x_0, e_\alpha\rangle e_\alpha, e_\beta \right\rangle = \langle x_0, e_\beta\rangle - \langle x_0, e_\beta\rangle = 0$$

$$\Rightarrow \quad x_0 - \sum_{\alpha \in I} \langle x_0, e_\alpha \rangle e_\alpha \perp \{e_\alpha\}_{\alpha \in I}$$

与 $\{e_\alpha\}_{\alpha \in I}$ 完备矛盾

Thm. 非平凡的 Hilbert 空间一定有 O.N.B.

例: l^2

$$e_n = (0, \cdots, 0, 1, 0, \cdots), \quad n = 1, 2, \cdots$$

($n-1$ 个 0)

$\{e_n\}_{n=1}^{\infty}$ 是 l^2 的 O.N.B.

($\because \quad (\{e_n\}_{n=1}^{\infty})^{\perp} = \{0\}$)

Rmk: $\{e_n\}_{n=1}^{\infty}$ 不是 l^2 的 Hamel basis.

$$(1, \frac{1}{2}, \frac{1}{3}, \cdots) \notin \operatorname{span}(\{e_n\}_{n=1}^{\infty})$$

Thm (Gram-Schmidt 正交化)

$(X, \langle \cdot, \cdot \rangle)$

$\{x_n\}_{n=1}^{\infty} \subset X$ 线性无关.

$\Rightarrow \exists \{e_n\}_{n=1}^{\infty}$ O.N.S. s.t. $\forall n$,

$$\operatorname{span}(\{e_k\}_{k=1}^{n}) = \operatorname{span}(\{x_k\}_{k=1}^{n})$$

Pf. /ℵ

$$y_1 = x_1 \qquad e_1 = \frac{y_1}{\|y_1\|}$$

$$y_2 = x_2 - \langle x_2, e_1 \rangle e_1 \qquad e_2 = \frac{y_2}{\|y_2\|}$$

$$\vdots \qquad \vdots$$

$$y_n = x_n - \sum_{k=1}^{n-1} \langle x_n, e_k \rangle e_k \qquad e_n = \frac{y_n}{\|y_n\|}$$

$$\vdots$$

$\forall\, m \neq n$. 不妨 $m < n$.

$$\langle e_m, e_n \rangle = \frac{1}{\|y_n\|} \langle e_m, x_n - \sum_{K=1}^{n-1} \langle x_n, e_K \rangle e_k \rangle$$

$$= \frac{1}{\|y_n\|} \left(\langle e_m, x_n \rangle - \overline{\langle x_n, e_m \rangle} \right)$$

$$= 0$$

由 e_j 的构造. $e_j \in \operatorname{span}\{x_i\}_{i=1}^{j}$

$\Rightarrow$ $\exists\, \alpha_{kj} \in \mathbb{K}$ s.t.

$$y_k = x_k - \sum_{j=1}^{k-1} \langle x_k, e_j \rangle e_j = \sum_{j=1}^{k-1} \alpha_{kj} x_j + x_k$$

$$\Rightarrow \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & & & \\ \alpha_{21} & 1 & & \\ \vdots & & \ddots & \\ \alpha_{n1} & \alpha_{n2} & \cdots & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$\Rightarrow \quad \operatorname{span}(\{y_k\}_{k=1}^{\wedge}) = \operatorname{span}(\{x_k\}_{k=1}^{n})$$

Thm H —— Hilbert space

H 可分 $\Leftrightarrow$ H 有至多可数 inf O.N.B.

Pf: 1° "$\Rightarrow$"

Case 1: $\dim H < \infty$

Hamel basis $\xrightarrow{G\text{-}S}$ O.N.B.

Case 2 $\dim H = \infty$

H 可分 $\Rightarrow \exists\ A = \{x_n\}_{n=1}^{\infty} \overset{\text{dense}}{\subset} H$

Claim. $\exists\ B \subset A$ 线性无关 s.t.

$$\operatorname{span}(B) = \operatorname{span}(A)$$

令 $n_1 = 1$

n_2 记录第一个 $x_{n_2} \in A \setminus \operatorname{span}\{x_{n_1}\}$ inf 的指标

$\vdots$

n_k 记录第一个 $x_{n_k} \in A \setminus \operatorname{span}\{x_{n_1}, \cdots, x_{n_{k-1}}\}$ inf 的指标.

よって $y_k = x_{n_k}$

$B = \{y_k\}_{k=1}^{\infty}$

$\Rightarrow$ $\forall x_k \in A$

$x_k \in \mathrm{span}\{y_1, \cdots, y_{k-1}, y_k\}$

(∵ x_{n_k} の取り方より $k \leq n_k$)

$\Rightarrow$ $A \subset \mathrm{span}(B)$

$\overline{A} = H$

$\Rightarrow$ $\overline{\mathrm{span}(B)} = \overline{\mathrm{span}(A)} = H$

$\Rightarrow$ $\#B = \infty$

$\dim \mathrm{span}(B) < \infty$ とすると

$\Rightarrow$ $\mathrm{span}(B)$ は閉集合

$\Rightarrow$ $\overline{\mathrm{span}(B)} = \mathrm{span}(B)$ より有限次元部分空間

$\Rightarrow \overline{\mathrm{span}(B)} = H$ 矛盾

$B \xrightarrow{G\text{-}S} \{e_n\}_{n=1}^{\infty}$

$\Rightarrow$ $\{e_n\}_{n=1}^{\infty}$ は O.N.S. s.t.

$\mathrm{span}(\{e_n\}_{n=1}^{\infty}) = \mathrm{span}(B) = \mathrm{span}(A)$

$\Rightarrow$ $\overline{\mathrm{span}\{e_n\}_{n=1}^{\infty}} = \overline{\mathrm{span}(A)} = H$

$$\Longleftrightarrow \quad \left(\{e_n\}_{n=1}^{\infty}\right)^{\perp} = \{0\}$$

$$\Longleftrightarrow \quad \{e_n\}_{n=1}^{\infty} \overset{n}{\subset} \text{O.N.B.}$$

HW: Ex. 1.6.11 – 12. 1.6.16.